Shayan Hajhashemi

World models · Representation learning · Dynamical systems and control

McGill University · Montréal, Canada Mila profile

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Research profile

Scientific machine learning researcher working on representation learning and nonlinear dynamical systems. Research spans multimodal world models, learned inverse solvers, and constrained optimal control in differentiable simulations. Interested in action-conditioned representations and reliable world models for robot learning and planning. Seeking research internships and full-time research roles.

Education

McGill University — PhD, Quantitative Life Sciences Expected 2026

Supervisors: Morgan Craig and Amin Emad. Developing scientific machine learning tools in nonlinear dynamics and control with applications in pharmacology.

McGill University — BSc (Honours), Interdepartmental Immunology 2021

Honours research: Investigating Turing patterns in reaction–diffusion systems; supervisor: Anmar Khadra.

Selected research

Multimodal world models: alignment, prediction, and control *AISTATS 2027 submission*

- Co-developed concepts for representation alignment and nonlinear model predictive control (NMPC) in multimodal world models; contributed to theoretical analysis.
- Benchmarked our regularizer of learned representations to SINDy and time-series foundation models (TSFMs) on the same NMPC task for the Lorenz system.

WeLD: label-free weak-form distillation for PDE inference *Research draft*

- Coauthored a label-free method that learns weights over weak-form equations, distilling classical estimates from long observation windows into PDE parameter inference from short windows.
- Analyzed teacher bias and instrumental-variable corrections; explored forecasting-based latent world models that learn parameter inference without a teacher.

WAR: weak-form adaptive regularization for system identification *AISTATS 2027 submission*

- Combined learned temporal-window selection and adaptive regularization with differentiable constrained weak-form solvers; used variable projection for nonlinear parameter inference from noisy observations.
- Showcased superior inverse problem performance on held-out systems across six ODE families, evaluating parameter and trajectory recovery separately. Tested sensitivity to noise and sparse or irregular sampling.

OIL: learning initializers through downstream optimization *ICLR 2027 submission*

- Trained message-passing initializers through five differentiable trajectory-matching updates, with a three-term loss on initial parameters, refined parameters, and simulated trajectories.
- Analyzed sensitivity geometry with Fisher information, investigated rollout failures and posterior diagnostics.

DiffDose: differentiable optimal control for personalized medicine

Preprint; under review at npj Systems Biology and Applications

- Optimized bounded dose amplitudes and administration times through differentiable ODE and delay differential equation simulations. Compared automatic differentiation with analytic adjoints and finite differences for open-loop control.

Latent-space dynamics and representation learning *Preprint; under review at Nature Computational Science*

- Coauthored a variational framework that constrains learned cell-state representations of single-cell RNA-seq data to flows on an energy landscape, using potential-gradient neural ODEs for trajectory and cell-fate prediction.

Publications and manuscripts

1. **Hajhashemi, S.**; Shum-Tim, C.; Craig, M.; Emad, A. *WAR: Weak-form Adaptive Regularization for Dynamical Systems Inverse Problems.* Submitted 7 October 2026. Under review at *AISTATS 2027*.

2. **Hajhashemi, S.**; Shum-Tim, C.; Prakash, S.; Emad, A. *Predators, Prey, and Sloppy Parameters: Fine-Tuning Amortized Inference for Generalized Lotka–Volterra.* Submitted 18 September 2026. Under review at *ICLR 2027*. **Co-first author.**

3. Ramezanian-Panahi, M.; **Hajhashemi, S.**; Rish, I. *Multimodal World Models: Alignment, Prediction, and Control.* Submitted 7 October 2026. Under review at *AISTATS 2027*.

4. **Hajhashemi, S.**; Emad, A.; Craig, M. *DiffDose: Differentiable Programming for Personalized Dose-Regimen Optimal Control.* Submitted 13 September 2026. bioRxiv DOI: 10.64898/2026.09.07.749974. Under review at *npj Systems Biology and Applications*.

5. Poursina, A.; **Hajhashemi, S.**; Mikaeili Namini, A.; Saberi, A.; Emad, A.; Najafabadi, H. S. *A Latent Space Thermodynamic Model of Cell Differentiation.* Submitted 6 March 2026. bioRxiv DOI: 10.64898/2026.03.04.709512. Under review at *Nature Computational Science*.

6. Lemaire, V.; **Hajhashemi, S.**; Rotte, A.; Seyed-Ahmadi, A.; Tripathi, S.; Chanu, P.; Wu, B.; Li, C.; Ramanujan, S.; Gadkar, K.; Jin, J. *Explainability analysis of QSP-derived virtual populations identifies robust pre-treatment biomarkers of immunotherapy.* Accepted for publication in *npj Precision Oncology*.

Auxiliary publications

7. Brassard, A.; Dhoparee-Doomah, I.; **Hajhashemi, S.**; Bourdeau, F.; Giannias, B.; Spicer, J.; Ferri, L. E.; Ma, K.; Cools-Lartigue, J. *Neutrophil Extracellular Trap Deposition Establishes an Immunosuppressive Environment in Non-metastatic Lymph Nodes and Facilitates Heterologous Metastatic Outgrowth.* 28 October 2026. bioRxiv, version 1. DOI: 10.64898/2026.10.23.761387.

8. Dhoparee-Doomah, I.; Ma, K.; Brassard, A.; Leo, S.; **Hajhashemi, S.**; …; Cools-Lartigue, J. *RORγ Inverse Agonism Redirects Tumour-Conditioned Emergency Myelopoiesis and Attenuates Established Tumour Progression.* 7 October 2026. bioRxiv, version 1. DOI: 10.64898/2026.10.01.755741. Under review at *Cancer Discovery*.

9. Dhoparee-Doomah, I.; Li, Y.; Al Dow, M.; **Hajhashemi, S.**; …; Cools-Lartigue, J. *A Measurable Systemic Immune Phenotype Links Circulating Myeloid Dysfunction to Tumour Spatial Organization in Gastroesophageal Adenocarcinoma.* 18 September 2026. bioRxiv, version 1. DOI: 10.64898/2026.09.17.752366.

10. Roche, P.; **Hajhashemi, S.**; Akavia, U. D. (2020). *Nanobody administration of Cas9 improves nuclear localization and HDR-enhanced synthetic lethality in A549 lung carcinoma.* Withdrawn for intellectual property reasons. Jenthera Therapeutics.

Full author lists and patent: shayshay42.github.io/publications.html.

Research experience

Genentech — Modeling & Simulations Intern 2024–2025
Supervisor: Vincent Lemaire; San Francisco, USA.

- Trained conditional generative models using flow matching and optimal transport for simulator-derived virtual populations; applied learned distributions to biomarker analysis.

Earlier research: Consultant Research Scientist, Jenthera Therapeutics (2020–2021); Research Associate, Goodman Cancer Research Center (2019–2020). Experimental nanobody research and computer vision.

Selected technical projects

Robot manipulation in simulation — Prototype for robot learning in simulation, 2022–2023

- Built training and evaluation scripts for simulated manipulation with MuJoCo, robosuite, Gymnasium, and Stable-Baselines3; explored soft actor–critic and behavior cloning from demonstrations.

Representation learning for dynamical systems and graphs — McGill, 2021–2022

- Explored time-series embeddings using Transformers, LSTM autoencoders, Fourier and wavelet features for novelty search; studied relational graph convolutional embeddings and structural comparison of 3D RNA motifs.

Technical skills

Programming: Python, Julia, MATLAB; PyTorch, JAX/diffrax, Pyro, Julia SciML.
Methods: Neural ODEs, graph neural networks, system identification, differentiable optimization, variational inference, flow matching, sensitivity analysis, model predictive control.
Simulation projects: MuJoCo, robosuite, Gymnasium, Stable-Baselines3; reinforcement and imitation learning experiments.

Presentations

- **Hajhashemi, S.** *Real-MVP: Virtual Populations As Simulation-Based Conditional Neural Densities.* SIAM Conference on the Life Sciences, Cleveland, Ohio, 6 July 2026. Invited oral presentation at an international conference. MS11: Recent Advances in Digital Twins and Trial Simulations to Guide Clinical Decisions.
- Ramezanian-Panahi, M.; **Hajhashemi, S.**; Rish, I. *Cross-modal control in the Virtual Brain.* Poster presentation (4D), MAIN 2026 — Montreal AI and Neuroscience, HEC Montréal, Montréal, Canada, 30 October 2026.
- Shum-Tim, C.; **Hajhashemi, S.** *Weak-Form Distillation for Label-Free PDE Inverse Problems.* World Modeling for Physics Workshop, Aspen Center for Physics, Aspen, Colorado, 28 February–5 March 2027. **Accepted oral presentation; upcoming.**

Teaching and service

Teaching: Scientific Machine Learning and ODEs in Julia, workshop instructor (2023); graduate nonlinear dynamics and computational methods, teaching assistant (2022).
Service: Julia conference reviewer (2024).

Selected awards

NSERC CGS-D (2023–2026); FRQ-Santé doctoral scholarship (awarded 2023–2026); department three-minute thesis winner (2022); Dean's Multidisciplinary First Class Honours (2019–2021).